

Measuring Personality for Automatic Elicitation of Privacy Preferences

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The paper, authored by Mehrpouyan, Azpiazu, and Pera, explores how personality traits influence privacy preferences and proposes an approach to automate privacy settings using machine learning. It examines the Big-Five personality traits—openness, conscientiousness, extraversion, agreeableness, and neuroticism—to predict user privacy behaviors. Through a survey of 413 participants, the study identifies patterns in privacy preferences and evaluates the feasibility of using machine learning to automate privacy management. The results suggest that personality traits significantly impact privacy decisions and that machine learning models can enhance the accuracy of privacy settings while reducing user effort.

The authors provide a clear explanation of key terms such as privacy management, Big-Five personality traits, and communication privacy management theory. They outline how privacy preferences vary based on personality traits and emphasize the necessity of automating privacy settings to ease the burden on users. However, certain sections, particularly in data analysis, contain complex statistical discussions that could be more accessible with clearer explanations. For instance, the discussion on Pearson's correlation in Section III-B (p. 87) includes mathematical notations and formulas that could benefit from further elaboration for readers unfamiliar with statistical techniques.

The main findings of the study indicate that personality traits significantly influence privacy preferences, and machine learning techniques can be leveraged to predict user behaviors. The study demonstrates that personality-based privacy models outperform random selection in determining optimal privacy settings, suggesting a promising direction for automated privacy management. Table VI (p. 91) highlights how the majority class for certain privacy decisions remains high, implying that predictive models can be effective in determining privacy preferences. Additionally, Table VII (p. 92) shows that the use of a multinomial random forest classifier improves classification accuracy in several cases. Figure 5 (p. 94) further illustrates that individuals scoring high in extraversion are more likely to share information than those with high neuroticism, providing empirical support for the hypothesis that personality influences privacy behaviors. However, the effectiveness of machine learning models is not extensive compared to existing privacy-setting approaches.

The paper's primary strength is its interdisciplinary approach, combining psychology and computer science to address privacy challenges. The use of machine learning to predict privacy preferences is innovative, providing an automated mechanism to enhance user privacy. **Figure 3 (p. 88) and Figure 4 (p. 89)** illustrate strong correlations between personality traits and privacy preferences, supporting the research hypothesis. The authors also contribute a dataset for future studies, enhancing privacy-aware computing research. The **Westin Privacy Segmentation Index (Table II, p. 86)** effectively categorizes users based on privacy attitudes, adding depth to the analysis. Furthermore, **Table VI (p. 91) and Table VII (p. 92)** demonstrate that machine learning models improve classification accuracy in predicting privacy behaviors. The study's **Pearson's correlation analysis (Table IV, p. 87)** validates relationships between personality traits, reinforcing the predictive approach. Additionally, the **survey-based data collection (p. 85)** ensures findings are based on real-world privacy behaviors from 413 participants, strengthening the study's credibility.

One weakness of the paper is the lack of diversity in its sample population, which affects the generalizability of the findings. **Page 86** shows that **77% of respondents were Caucasian, with only 7% African American and 7% Hispanic**, limiting broader representation. **Figure 2 (p. 86)** reveals that a significant portion of participants are over 55 years old, potentially overlooking younger users' privacy preferences. Additionally, **Table III (p. 86)** shows that **64% of men and 66% of women prefer not to share information**, highlighting potential gender biases in the dataset. The study also focuses primarily on static characteristics like culture and gender but **does not adequately consider dynamic factors like evolving privacy behaviors (p. 85-86)**, which are critical for real-world applications. Lastly, **Table V (p. 90)** provides detailed privacy preferences but lacks contextual depth, making it difficult to determine whether responses reflect actual user intent or situational constraints.

In conclusion, the paper presents a valuable contribution to privacy-aware computing by demonstrating the feasibility of personality-based privacy prediction. While the study provides strong initial evidence for the use of machine learning in automating privacy settings, further work is needed to enhance model accuracy, incorporate contextual privacy factors, and validate findings with a more diverse participant base. Nonetheless, this research lays an important foundation for future developments in intelligent privacy management systems.